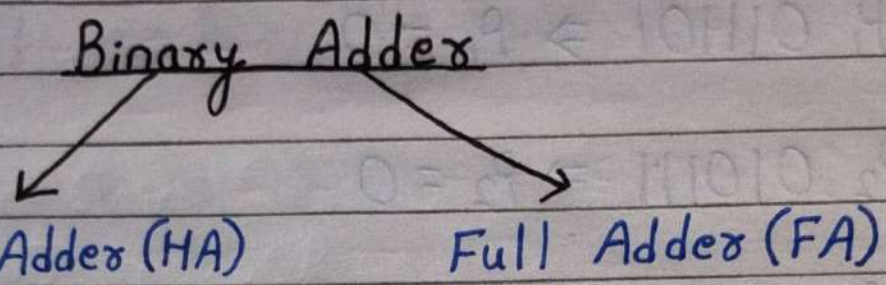
Module - II

Combinational Logic

Combinational Logic Circuit

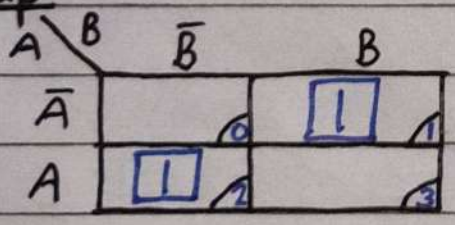


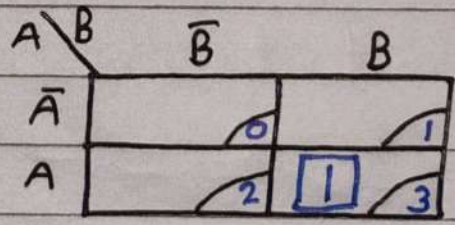
Half Adder

Truth Table:

	A	B	Sum (S)	Carry (C)
0	0	0	0	0
1	0	1	1	0
2	1	0	1	0
3	1	1	0	1

K-map:

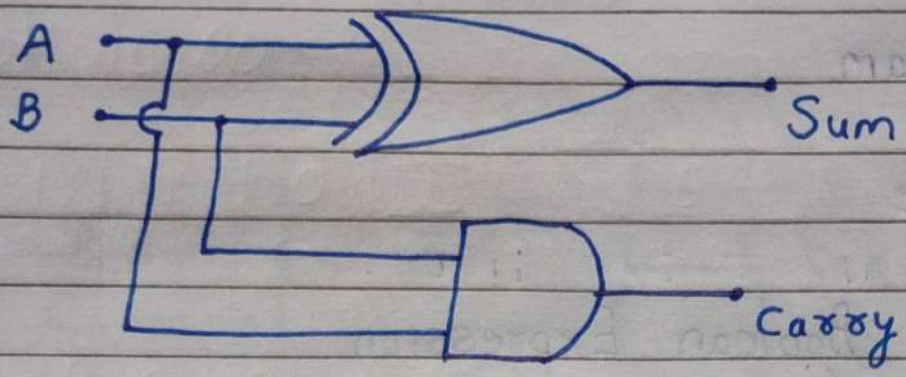




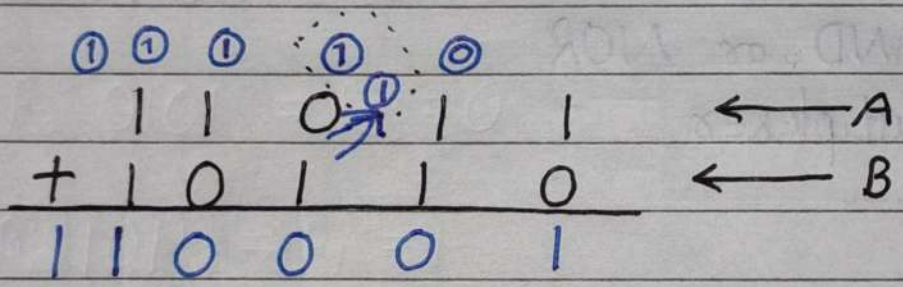
$Sum = \bar{A}B + A\bar{B}$ $\quad$ $Carry = AB$

$\Rightarrow Sum = A \oplus B$ [XOR] $\quad$ $\Rightarrow Carry = A.B$ [AND]

Logic Circuit:

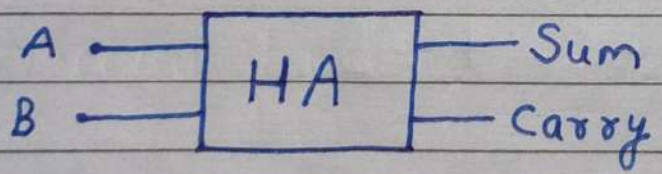


Example:



Here, if a carry is obtain then another half adder is required, which is the limitation.

Block Diagram:

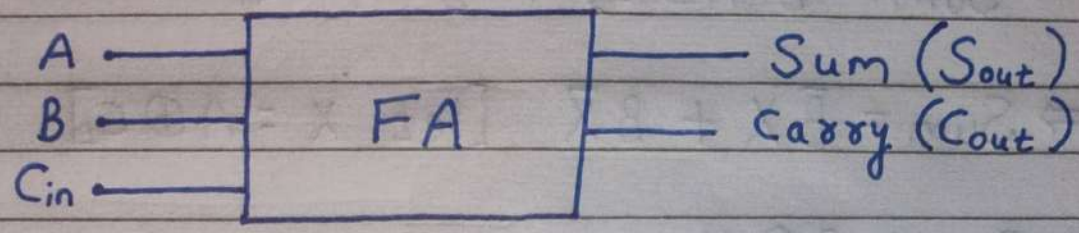


Design of Any Logic Circuit [Steps]

1. Block diagram
2. Truth table
3. K-map for Boolean Expression
4. Design
   - Basic Gate
   - NAND or NOR
   - Multiplexer

Full Adder

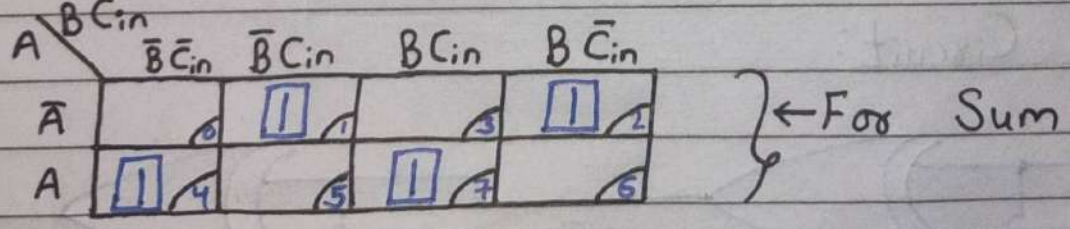
Block Diagram:



Truth Table:

	A	B	Cin	Sum	Carry
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

K-map:



$$Sum = A\bar{B}\bar{C}_{in} + \bar{A}\bar{B}C_{in} + ABC_{in} + \bar{A}B\bar{C}_{in}$$

$$\Rightarrow Sum = \bar{B}(A\bar{C}_{in} + \bar{A}C_{in}) + B(AC_{in} + \bar{A}\bar{C}_{in})$$

$\Rightarrow Sum = \bar{B}(A \oplus C_{in}) + B(A \odot C_{in})$

XOR X-NOR

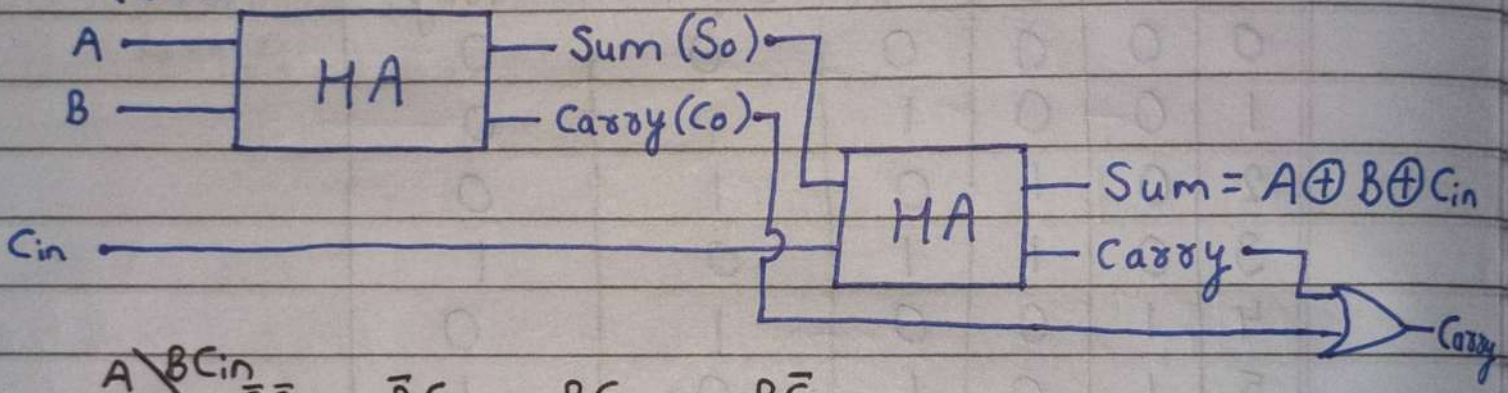
$\Rightarrow Sum = \bar{B}(A \oplus C_{in}) + B\overline{(A \oplus C_{in})}$

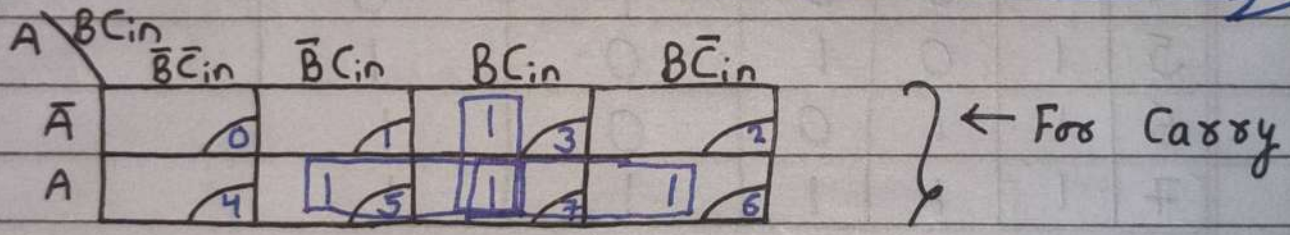
$\Rightarrow Sum = \bar{B}X + B\bar{X}$ [let $X = A \oplus C_{in}$]

$\Rightarrow Sum = B \oplus X$

$\Rightarrow Sum = A \oplus B \oplus C_{in}$

(S_{out})

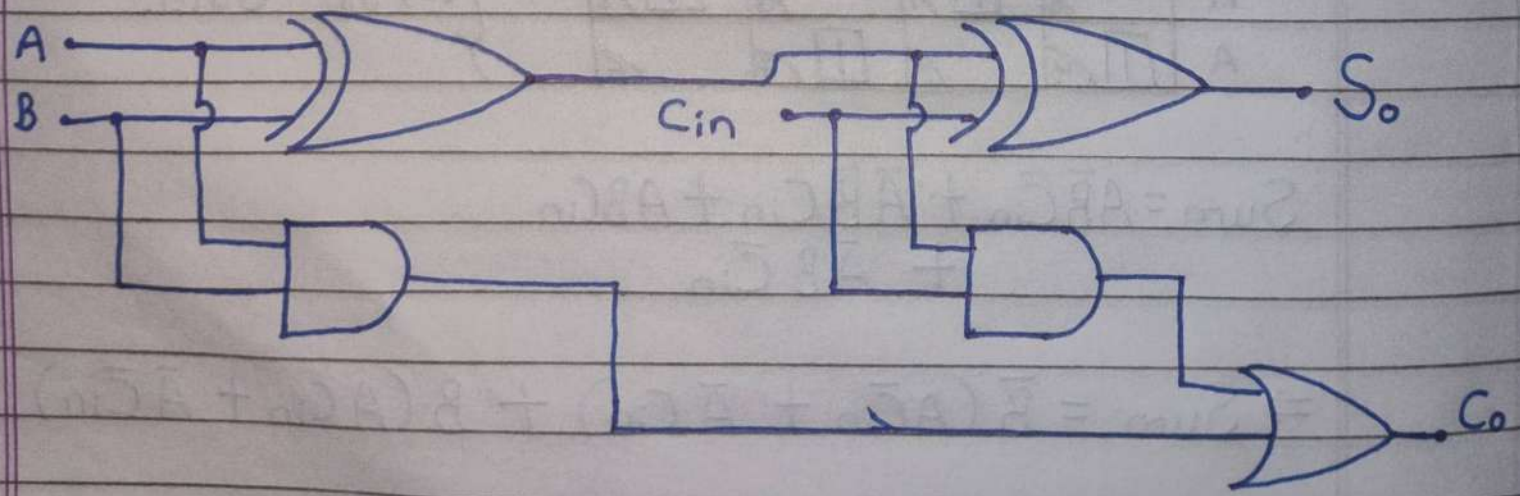




$Carry = BC_{in} + AC_{in} + AB.$

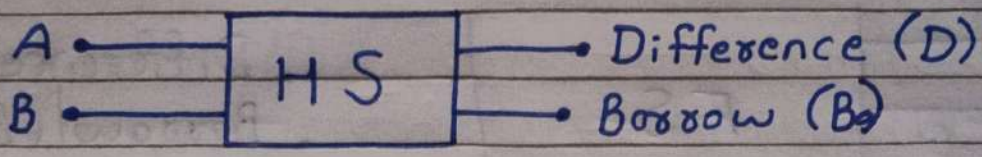
(C_{out})

Logic Circuit:



Half Subtractor

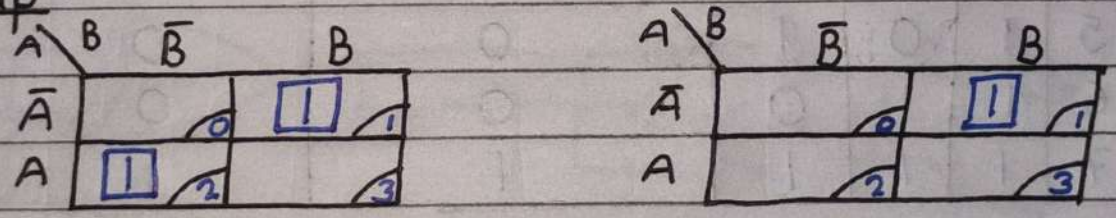
Block Diagram:



Truth Table:

	A	B	Difference (D)	Borrow (Bo)
0	0	0	0	0
1	0	1	1	1
2	1	0	1	0
3	1	1	0	0

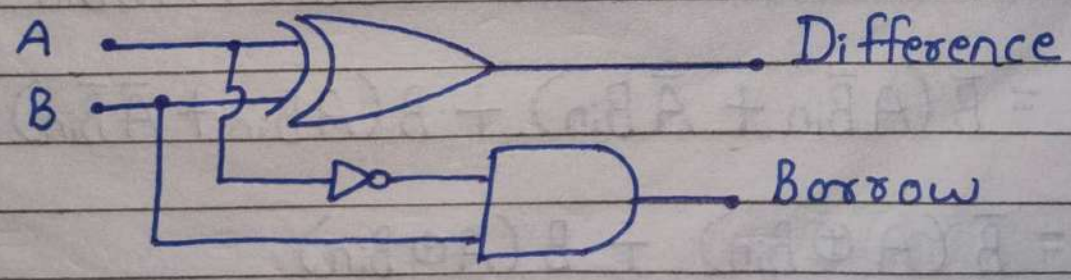
K-map:



$$\text{Difference} = A\bar{B} + \bar{A}B = A \oplus B \quad [\text{XOR}]$$

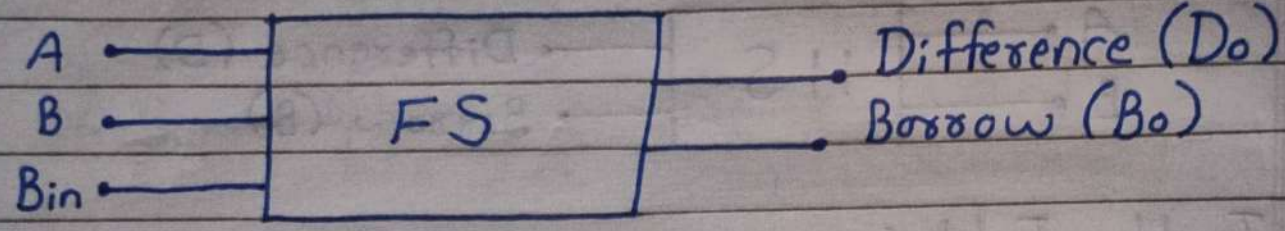
$$\text{Borrow} = \bar{A}B$$

Logic Circuit:



Full Subtractor

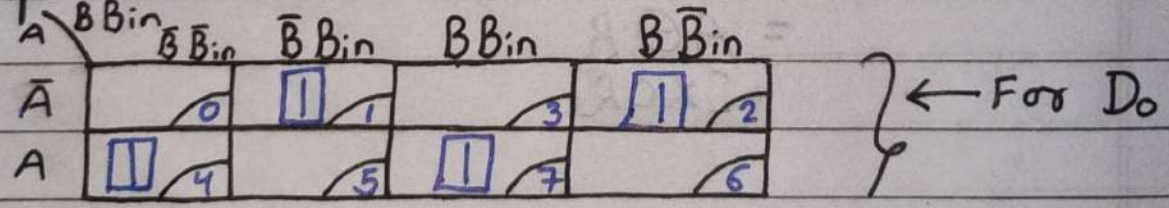
Block Diagram:



Truth Table:

	A	B	Bin	Difference (Do)	Borrow (Bo)
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

K-map:



$$\text{Difference } (D_o) = A\bar{B}\bar{B}_{in} + \bar{A}\bar{B}B_{in} + ABB_{in} + \bar{A}B\bar{B}_{in}$$

$$\Rightarrow D_o = \bar{B}(A\bar{B}_{in} + \bar{A}B_{in}) + B(AB_{in} + \bar{A}\bar{B}_{in})$$

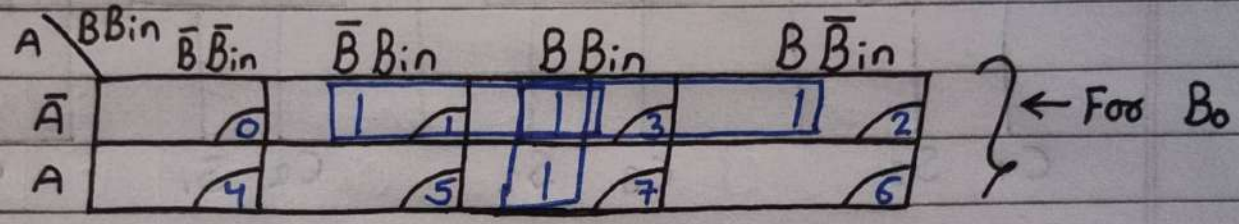
$$\Rightarrow D_o = \bar{B}(A \oplus B_{in}) + B(A \odot B_{in})$$

$\Rightarrow D_o = \bar{B}(A \oplus B_{in}) + B(\overline{A \oplus B_{in}})$

$\Rightarrow D_o = \bar{B}X + B\bar{X}$ [Let $X = A \oplus B_{in}$]

$\Rightarrow D_o = B \oplus X$

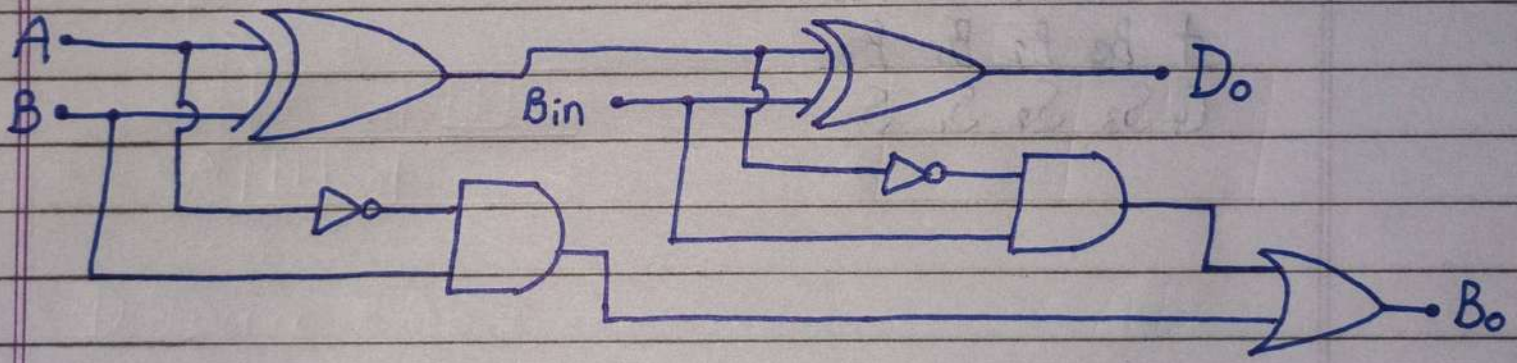
$\Rightarrow$ Difference $(D_o) = A \oplus B \oplus B_{in}$



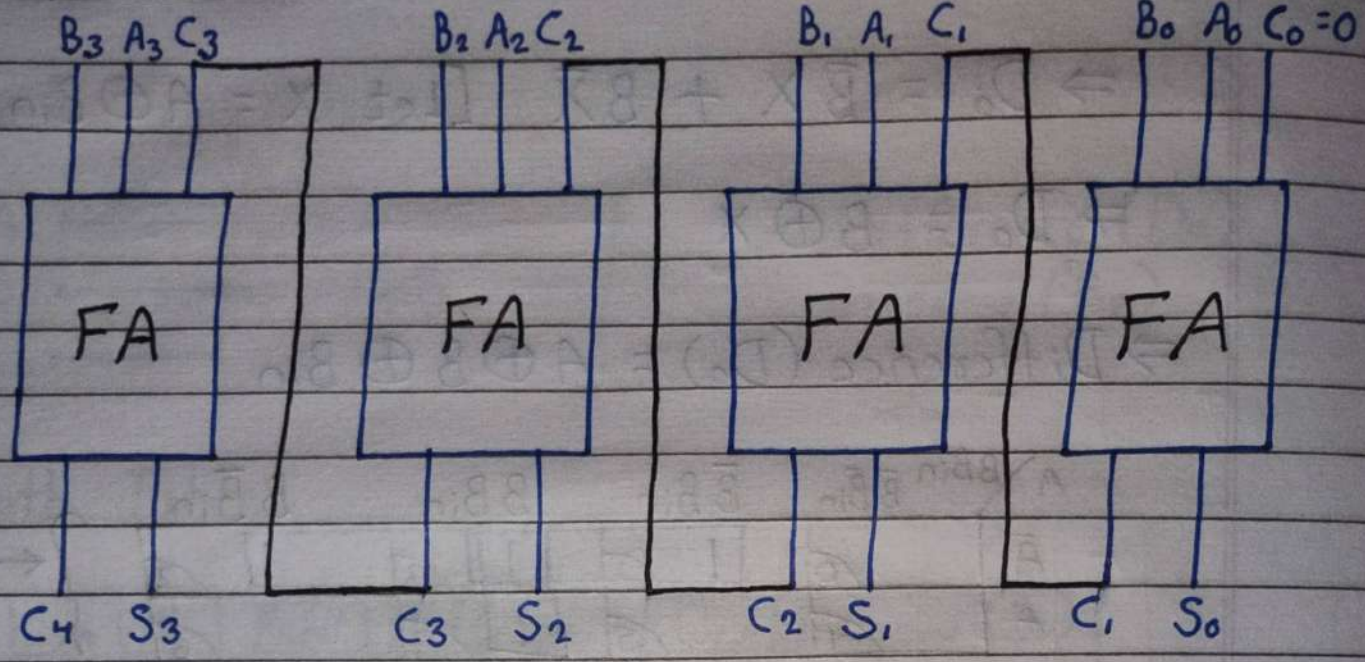
Borrow $(B_o) = \bar{A}B_{in} + BB_{in} + \bar{A}B$

or $B_o = \bar{A}B + B_{in}(\bar{A} + B)$

Logic Circuit:



Parallel Adder

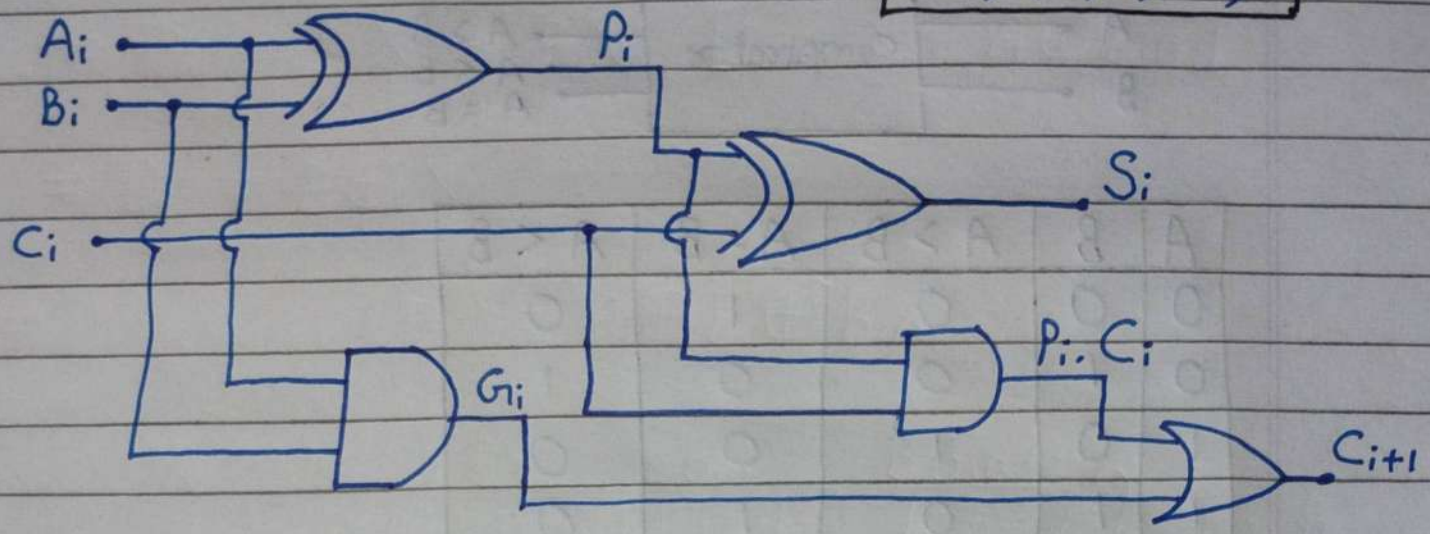


Final Sum: | C_4 | S_3 | S_2 | S_1 | S_0 |

$$
\begin{array}{r}
C_4\ C_3\ C_2\ C_1\ C_0 \\
A_3\ A_2\ A_1\ A_0 \\
+\ B_3\ B_2\ B_1\ B_0 \\
\hline
C_4\ S_3\ S_2\ S_1\ S_0
\end{array}
$$

Look Ahead Carry Adder

$i = 0, 1, 2, 3, \ldots, n$



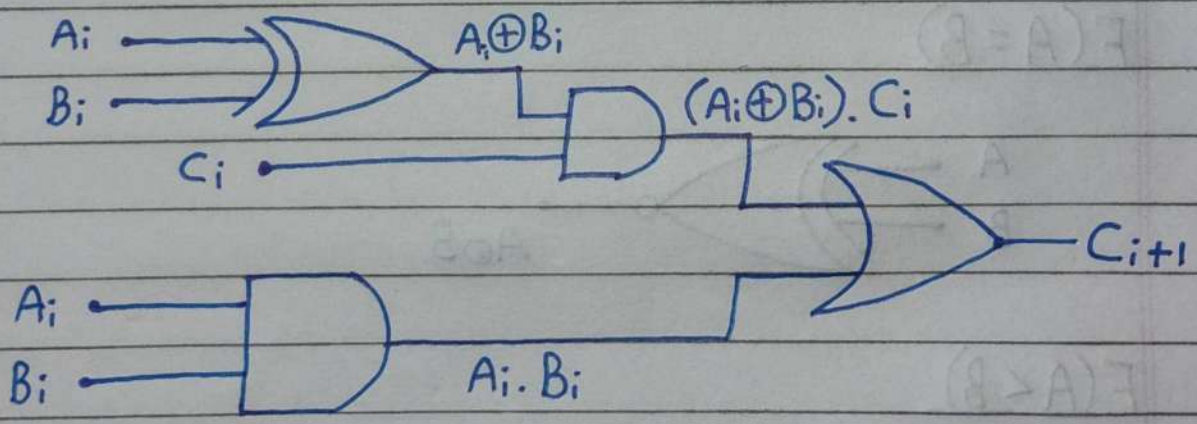
$P_i = A_i \oplus B_i$

$G_i = A_i . B_i$

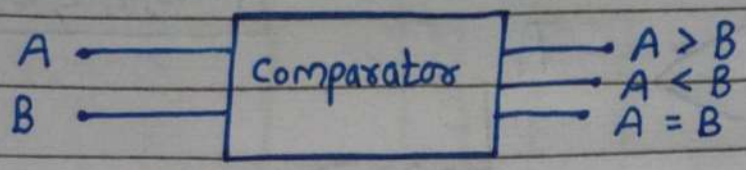
$S_i = P_i \oplus C_i = A_i \oplus B_i \oplus C_i$

$C_{i+1} = G_i + P_i . C_i$

$\Rightarrow \boxed{C_{i+1} = A_i . B_i + (A_i \oplus B_i) . C_i}$



Comparator



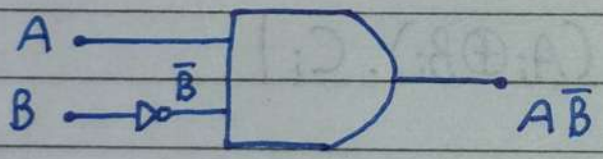
A	B	A > B	A = B	A < B
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

$F(A>B) = A\bar{B}$

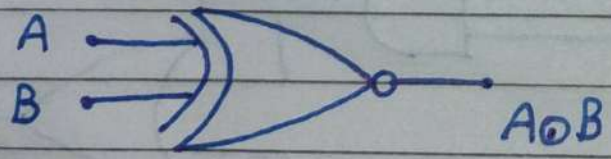
$F(A=B) = \bar{A}\bar{B} + AB = A \odot B$

$F(A<B) = \bar{A}B$

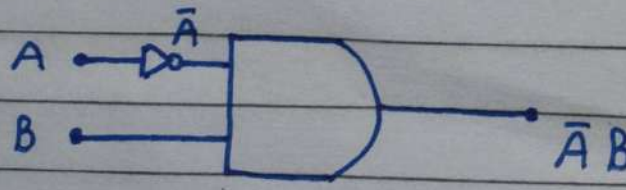
F(A > B)



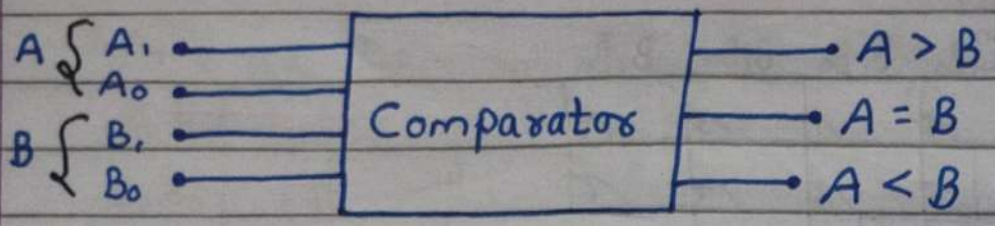
F(A = B)



F(A < B)

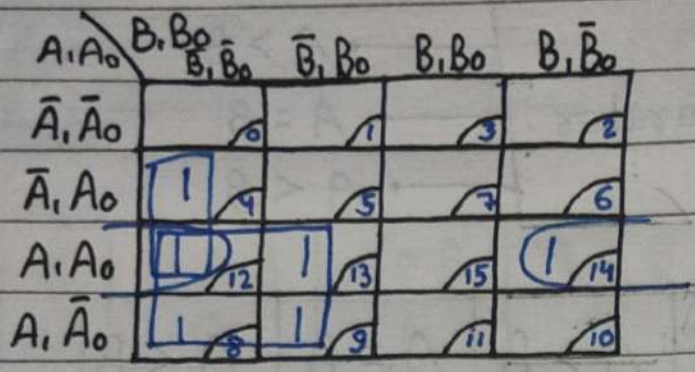


Comparator [2 Bit]



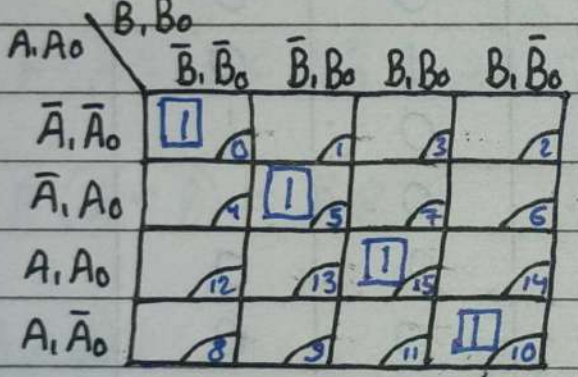
	A		B		A>B	A=B	A<B
	A_1	A_0	B_1	B_0			
0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1
4	0	1	0	0	1	0	0
5	0	1	0	1	0	1	0
6	0	1	1	0	0	0	1
7	0	1	1	1	0	0	1
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	1	0
11	1	0	1	1	0	0	1
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	1	0

K-map for A > B



$$F(A>B) = A_1\bar{B}_1 + A_0\bar{B}_1\bar{B}_0 + A_1A_0\bar{B}_0$$

K-map for A = B



$$F(A=B) = \bar{A}_1\bar{A}_0\bar{B}_1\bar{B}_0 + \bar{A}_1A_0\bar{B}_1B_0 + A_1A_0B_1B_0 + A_1\bar{A}_0B_1\bar{B}_0$$

$$= \bar{A}_1\bar{B}_1(\bar{A}_0\bar{B}_0 + A_0B_0) + A_1B_1(A_0B_0 + \bar{A}_0\bar{B}_0)$$

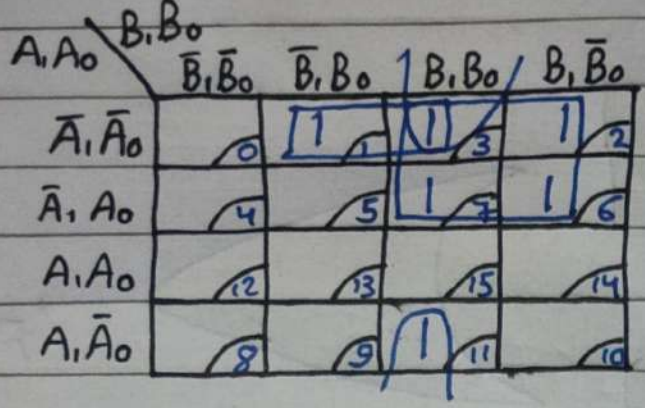
(X-NOR) (X-NOR)

$$= (A_0 \odot B_0)(\bar{A}_1\bar{B}_1 + A_1B_1)$$

(X-NOR)

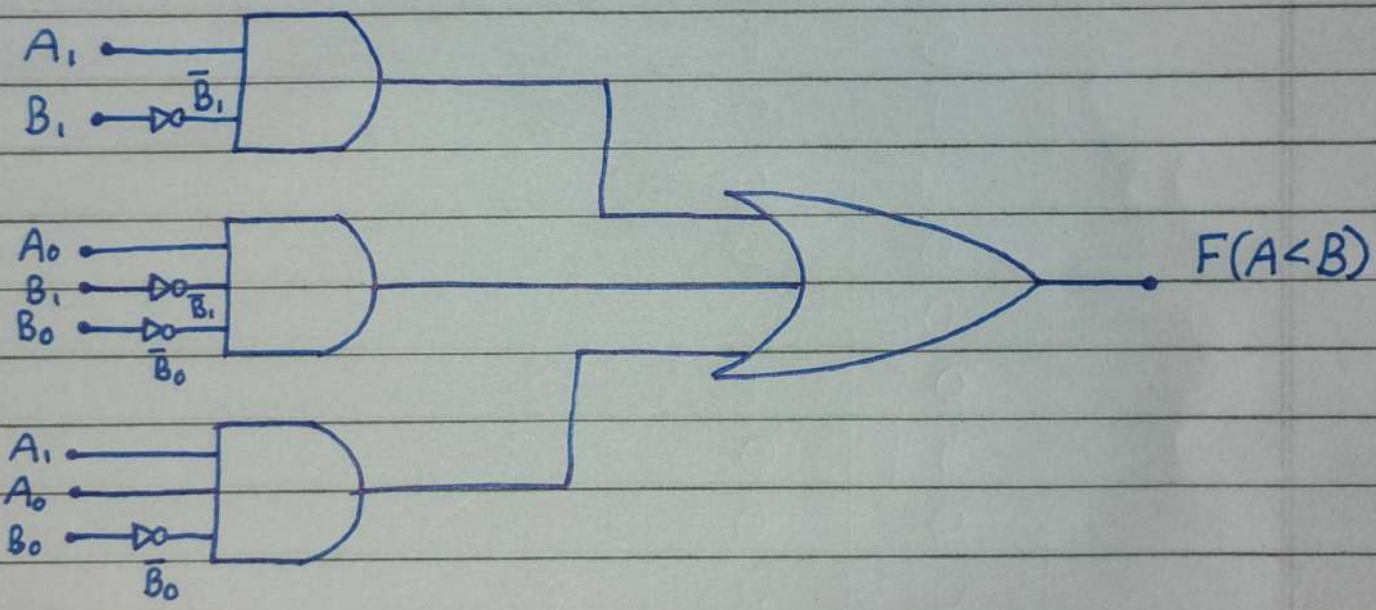
$$= (A_0 \odot B_0)(A_1 \odot B_1)$$

K-map for $A < B$

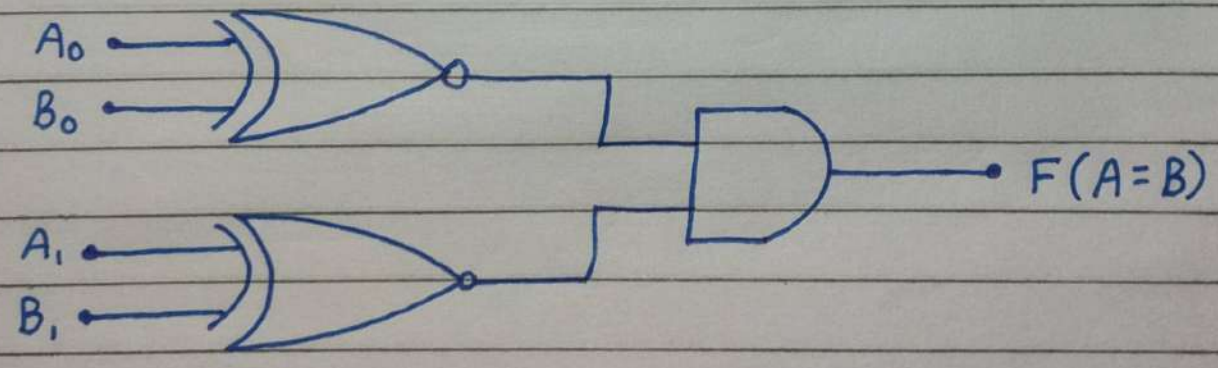


$$F(A<B) = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$

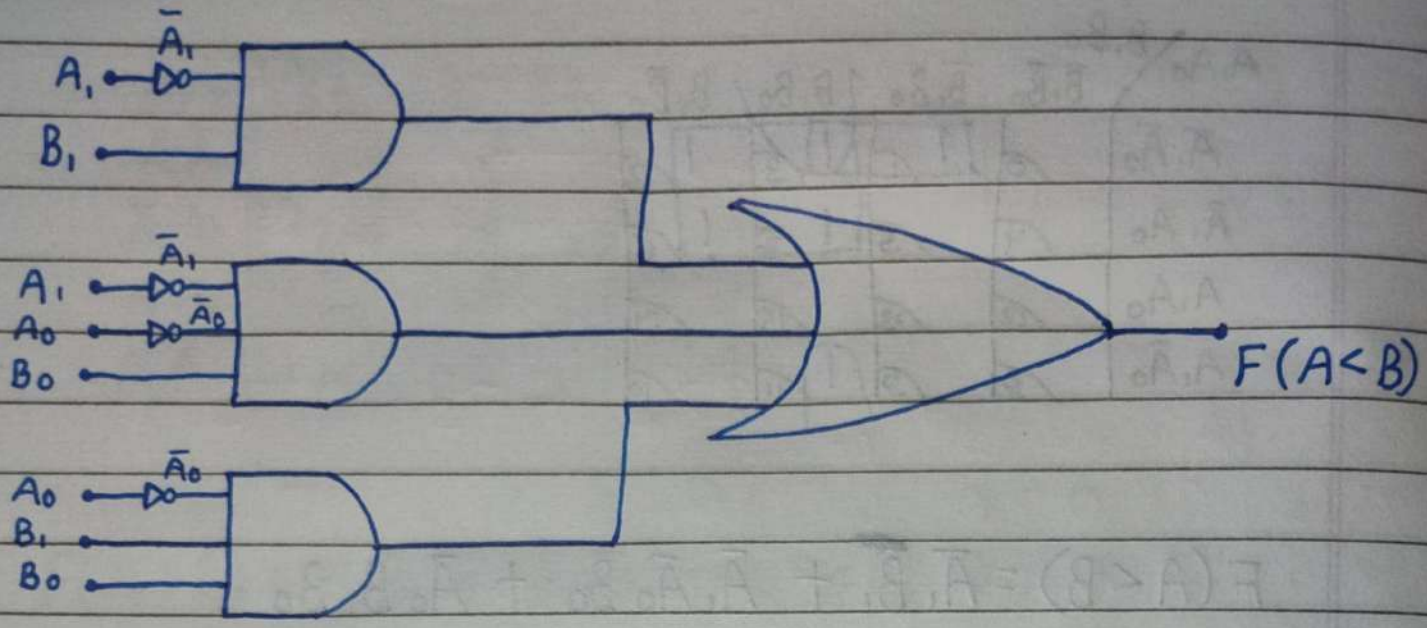
$A > B$



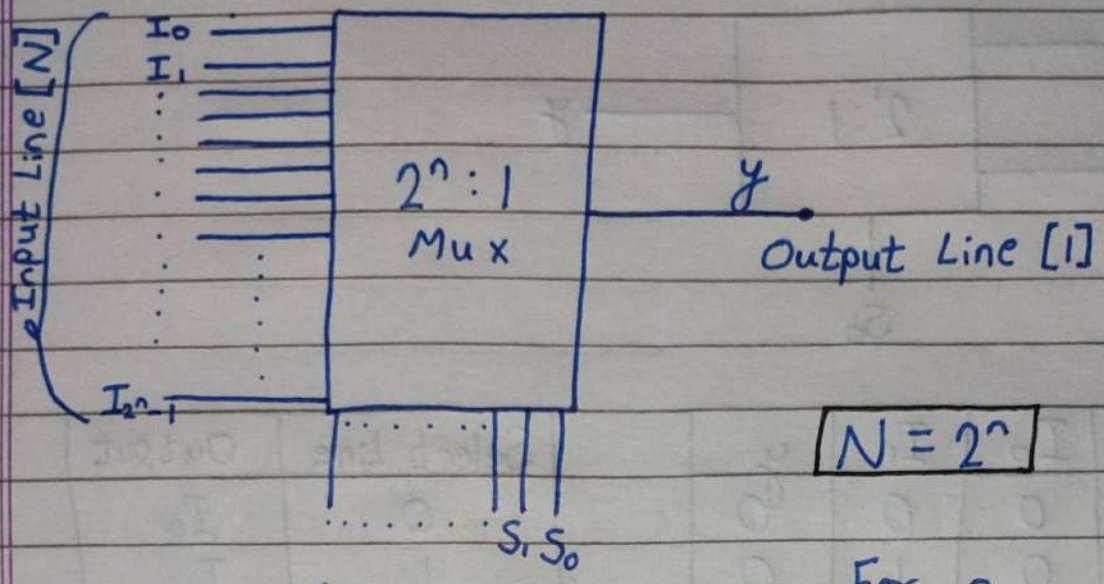
$A = B$



A < B



Multiplexer



$N = 2^n$

For n,

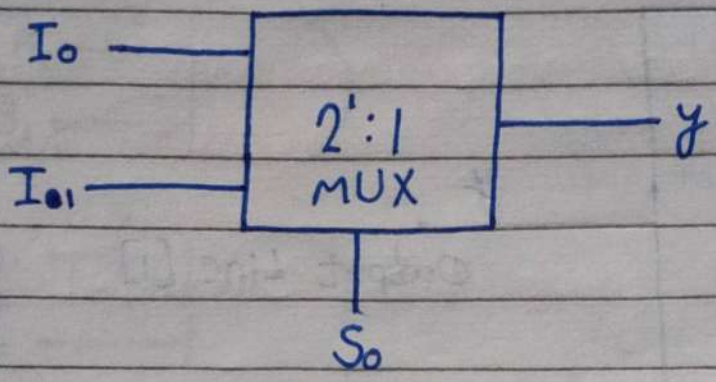
$\log_{10} N = n \log_{10} 2$

$n = \frac{\log_{10} N}{\log_{10} 2} = \log_2 N$

Uses:

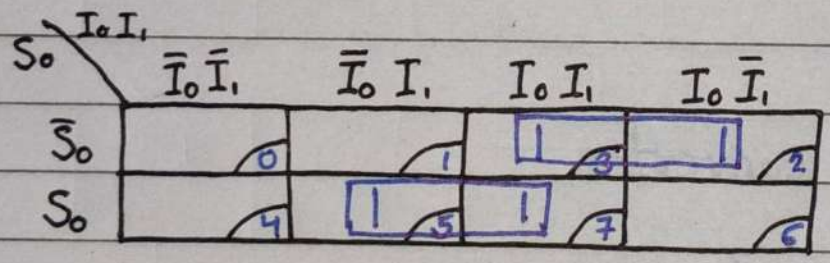
- Data selector & Data Routing
- Switch
- Many-to-one converter
- Parallel-to-serial converter
- Universal logic circuit (Implement any type of logic & gate.)

For 2:1 MUX

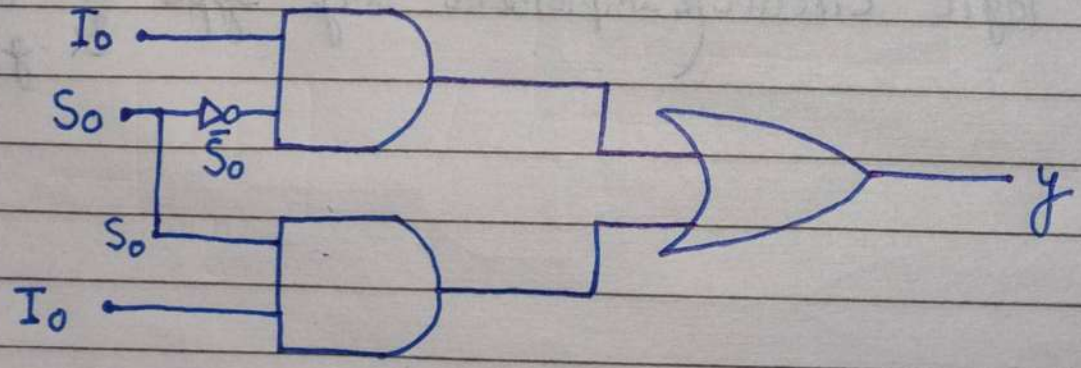


	S_0	I_0	I_1	y
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

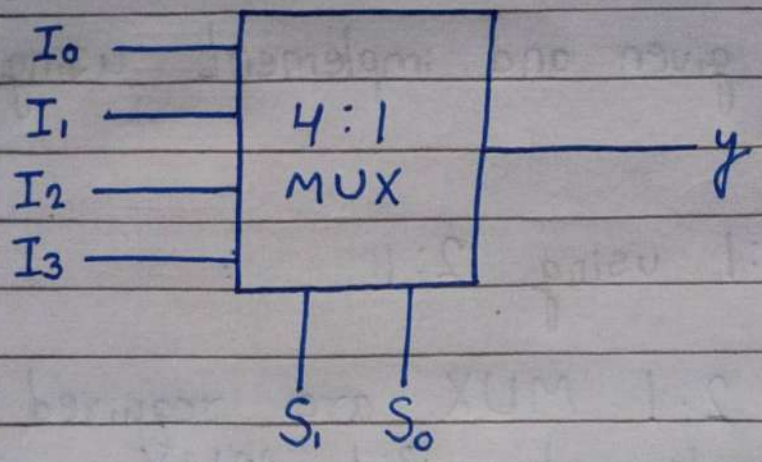
Select Line	Output
0	I_0
1	I_1



$$y = \bar{S}_0 I_0 + S_0 I_1.$$

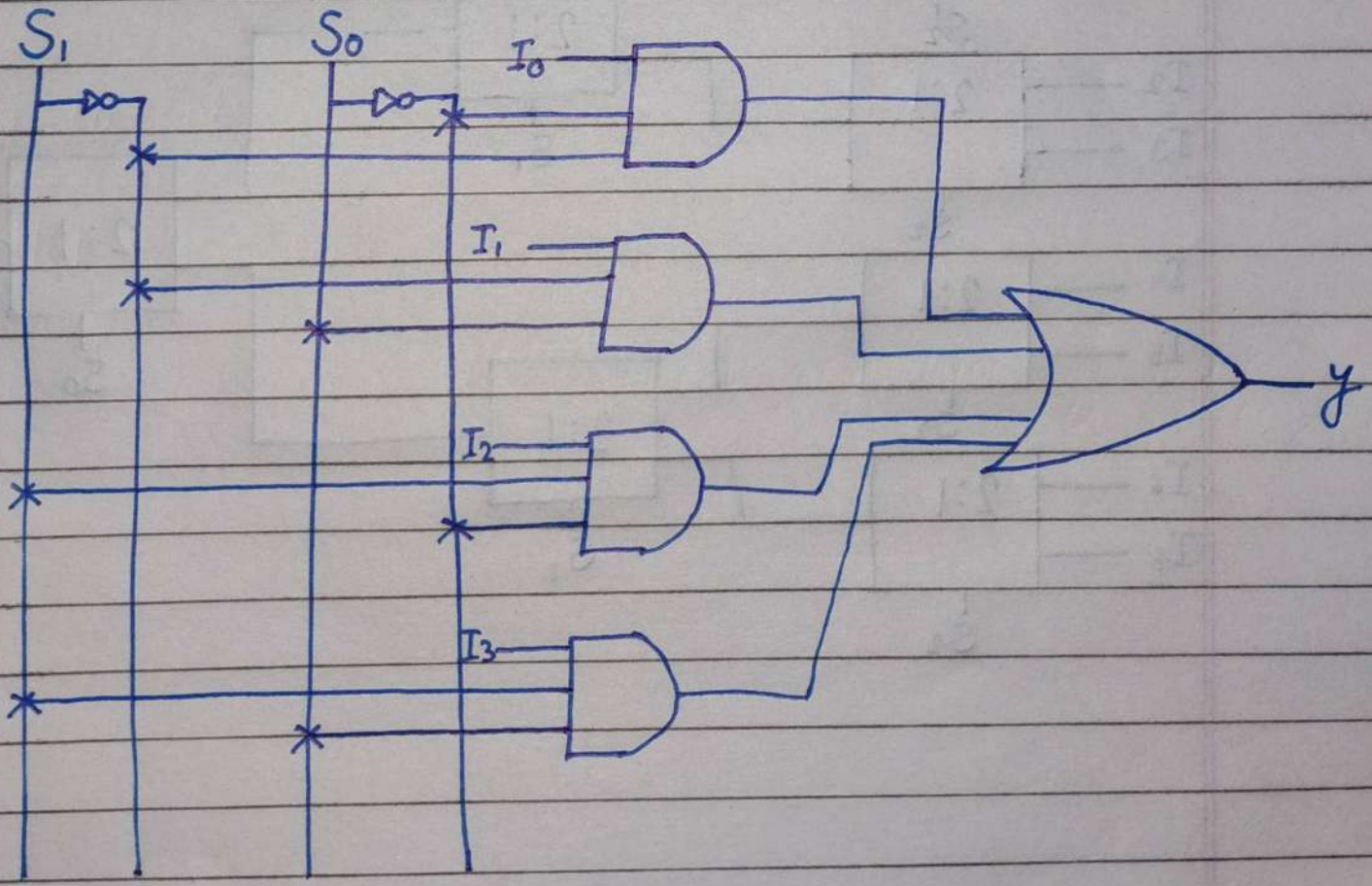


For 4:1 MUX



	S_1	S_0	y
0	0	0	I_0
1	0	1	I_1
2	1	0	I_2
3	1	1	I_3

$$y = \overline{S_1}\,\overline{S_0}\,I_0 + \overline{S_1}\,S_0 I_1 + S_1\,\overline{S_0}\,I_2 + S_1 S_0 I_3$$



Question Type - 1

Higher order are given and implement using lower order multiplexer.

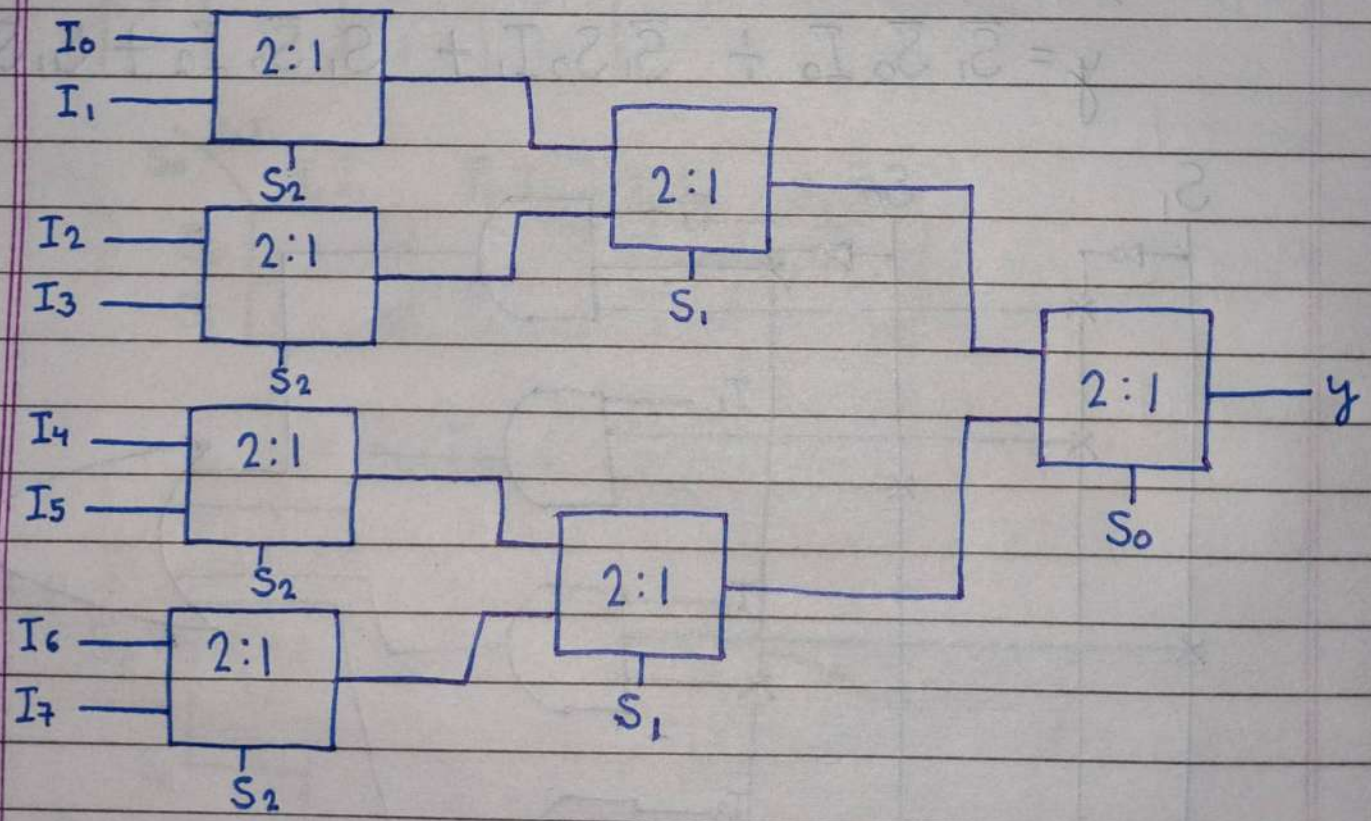
(Q) Implement 8:1 using 2:1

$\frac{8}{2} = 4$

$\frac{4}{2} = 2$

$\frac{2}{2} = 1$

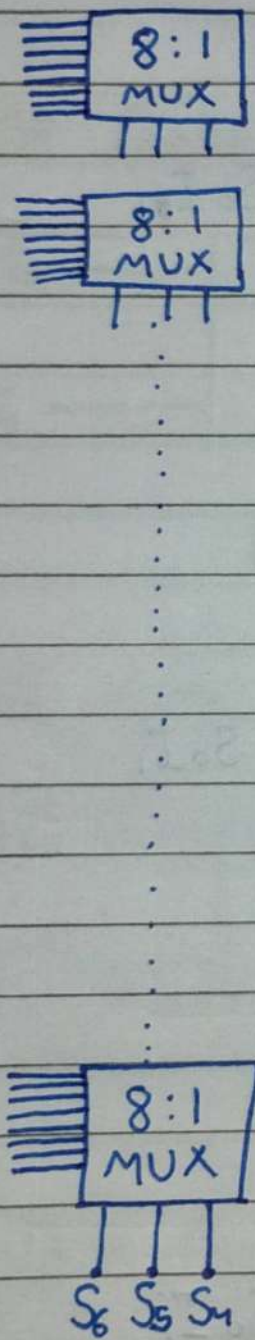
7 2:1 MUX are required to implement 8:1 MUX.



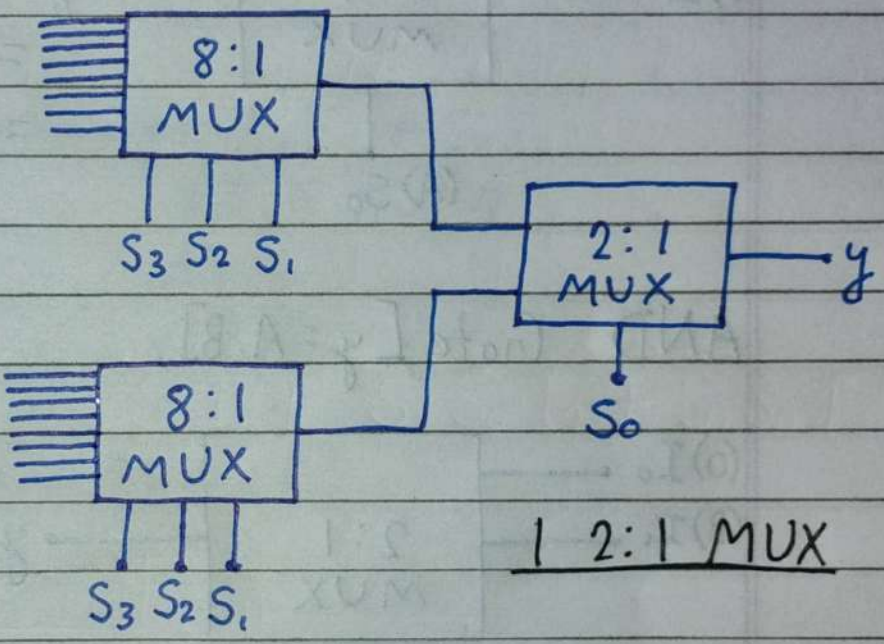
(8) Implement 128 : 1 using 8 : 1

$\frac{128}{8} = 16$

$\frac{16}{8} = 2$



16 8:1 MUX



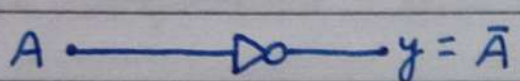
2 8:1 MUX

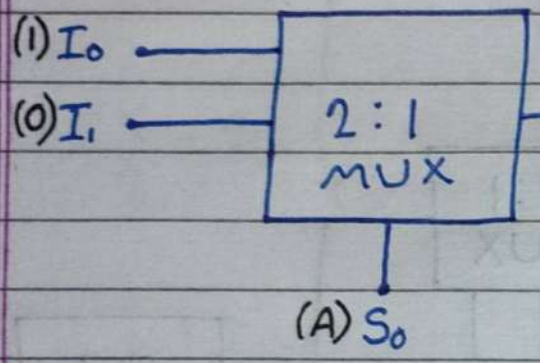
1 2:1 MUX

Question Type - 2

Multiplexer as a Logic Gate.

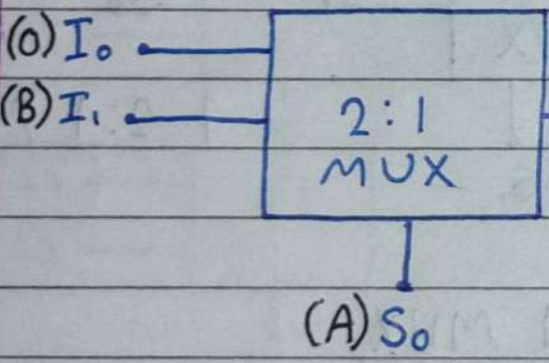
NOT Gate, $[y = \bar{A}]$,





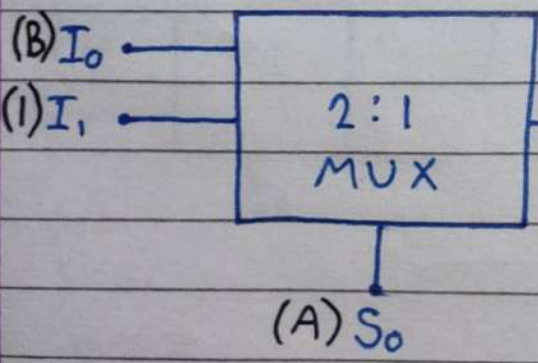
$y = \bar{S}_0 I_0 + S_0 I_1$
$= \bar{A} \cdot 1 + 0$
$= \bar{A}$

AND Gate $[y = A \cdot B]$,



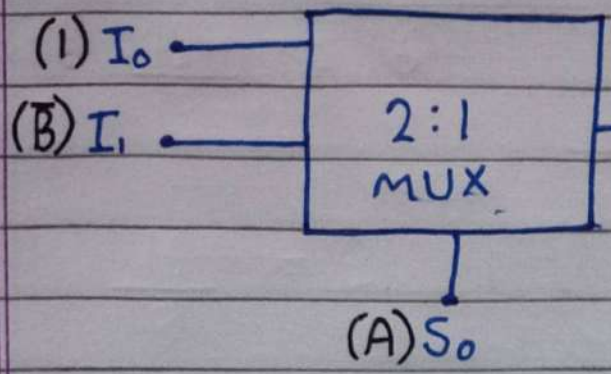
$y = \bar{S}_0 I_0 + S_0 I_1$
$= 0 + AB$
$= AB$

OR Gate, $[y = A + B]$,



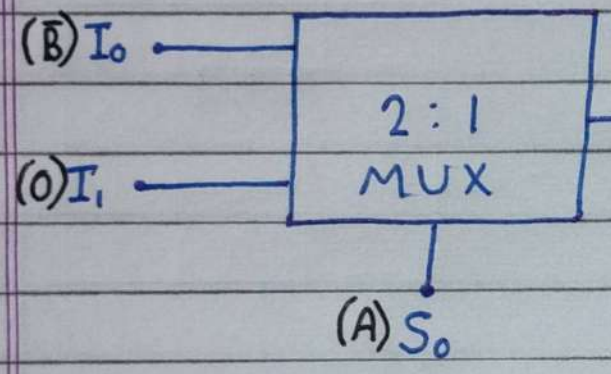
$y = \bar{S}_0 I_0 + S_0 I_1$
$= \bar{A} B + A \cdot 1$
$= (\bar{A} + A)(B + A)$
$= (A + B) \cdot 1$
$= (A + B)$

NAND Gate $[y = \overline{A.B}]$,



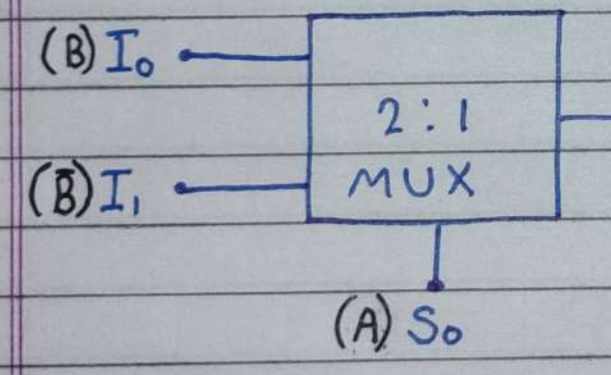
$$y = \bar{S}_0 I_0 + S_0 I_1$$
$$= \bar{A} \cdot 1 + A \cdot \bar{B}$$
$$= (\bar{A} + A)(\bar{A} + \bar{B})$$
$$= \bar{A} + \bar{B}$$
$$= \overline{AB}$$

NOR Gate $[y = \overline{A+B}]$,



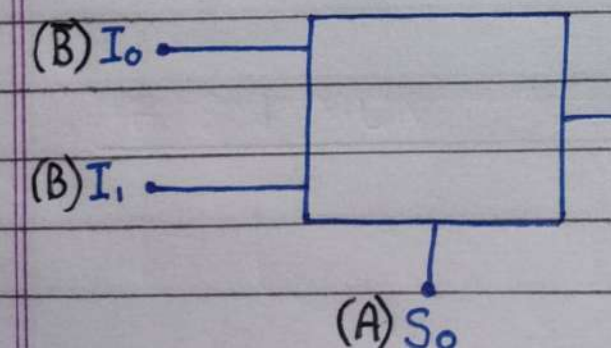
$$y = \bar{S}_0 I_0 + S_0 I_1$$
$$= \bar{A} \cdot \bar{B} + A \cdot 0$$
$$= \bar{A} \cdot \bar{B}$$
$$= \overline{A+B}$$

XOR Gate $[y = A \oplus B = \bar{A}B + A\bar{B}]$,



$$y = \bar{S}_0 I_0 + S_0 I_1$$
$$= \bar{A}B + A\bar{B}$$

X-NOR Gate $[y = A \odot B = AB + \bar{A}\bar{B}]$,

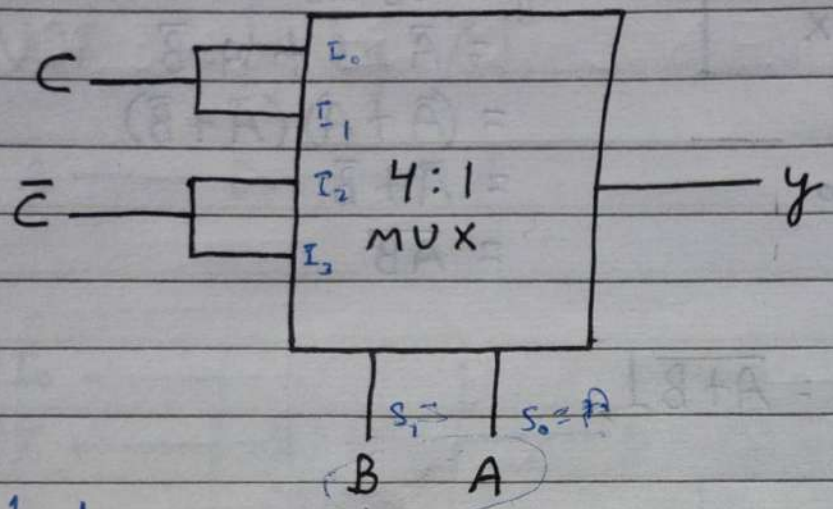


$$y = \bar{S}_0 I_0 + S_0 I_1$$
$$= \bar{A}.\bar{B} + A.B$$

Question Type -3

Minimization of given MUX.

Q.1)



We have,

$$y = \bar{S}_0 \bar{S}_1 I_0 + \bar{S}_0 S_1 I_1 + S_0 \bar{S}_1 I_2 + S_0 S_1 I_3$$

For the given MUX,

$$y = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + AB\bar{C}$$

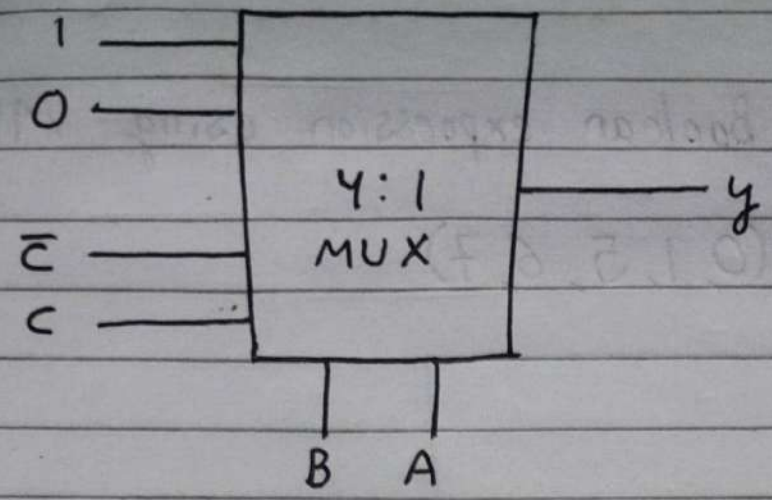
$$\Rightarrow y = \Sigma m(1, 3, 4, 6)$$

A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1 1	1 3	2
A	1 4	5	7	1 6

$$y = \bar{A}C + A\bar{C}$$

$$\Rightarrow y = A \oplus C$$

Q.2)



$$y = \bar{A}\bar{B}.1 + \bar{A}B.0 + A\bar{B}\bar{C} + ABC$$

$$\Rightarrow y = \bar{A}\bar{B} + B.0 + A\bar{B}\bar{C} + ABC$$

$$\Rightarrow y = \bar{A}\bar{B}(C+\bar{C}) + A\bar{B}\bar{C} + ABC$$

$$\Rightarrow y = \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + ABC$$

$$\Rightarrow y = \Sigma m\,(1, 0, 4, 7)$$

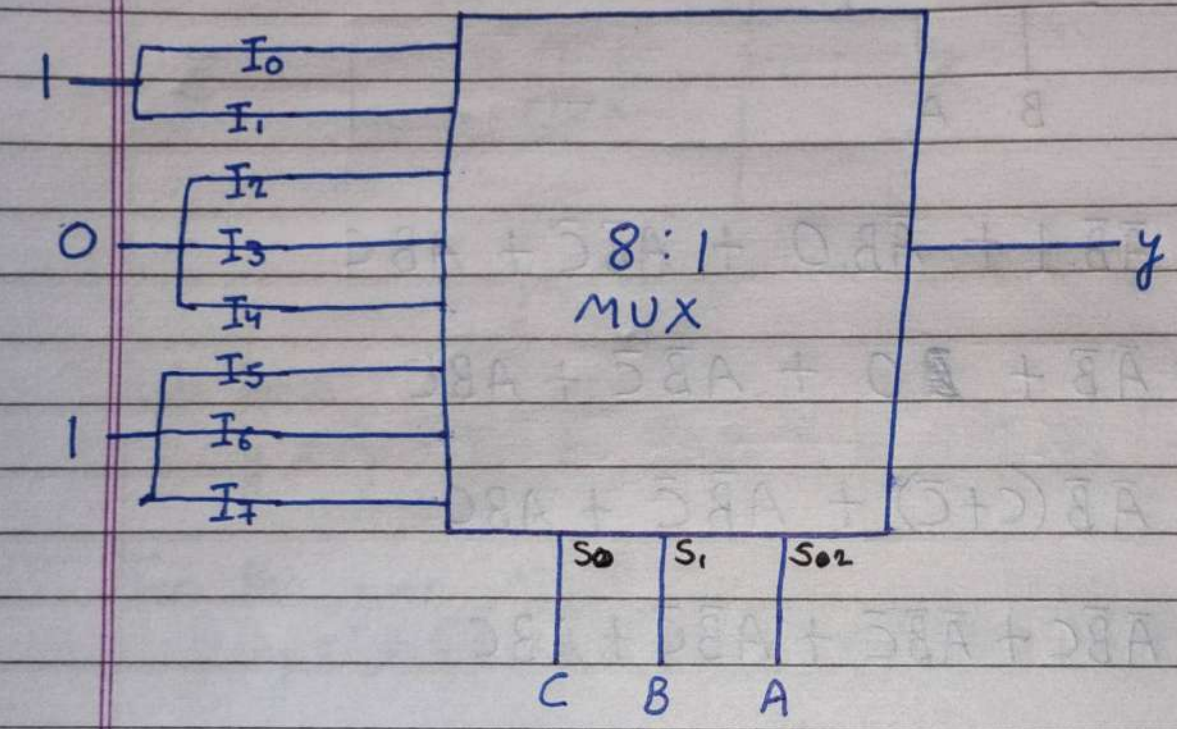
A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	1 (0)	1 (1)	3	2
A	1 (4)	5	1 (7)	6

$$y = \bar{A}\bar{B} + \bar{B}\bar{C} + ABC.$$

Question Type - 4

Implementation of Boolean expression using MUX.

Q. $y(A, B, C) = \Sigma_m(0, 1, 5, 6, 7)$



Q. Implement $y(A, B, C) = \Sigma_m(0, 1, 5, 6, 7)$ using 4:1 MUX.

Soln:

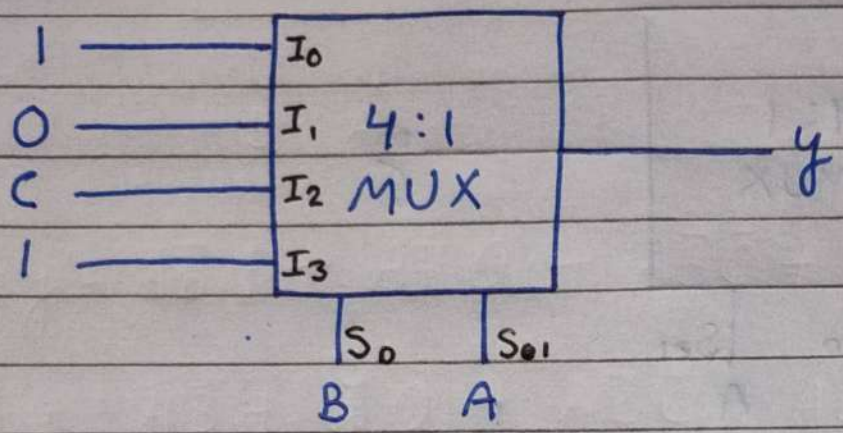
In 4:1 MUX, there are four input lines and 2 select lines but we have three variables. So we have to take select lines in the group of two and other one as input lines. The combinations are AB, BC & AC.

For A,B as select lines and C as input line,

	$\bar{A}\bar{B}$ (0)	$\bar{A}B$ (1)	$A\bar{B}$ (2)	AB (3)
$\bar{C}$	0	2	4	6
C	1	3	5	7
	1	0	C	1
	I_0	I_1	I_2	I_3

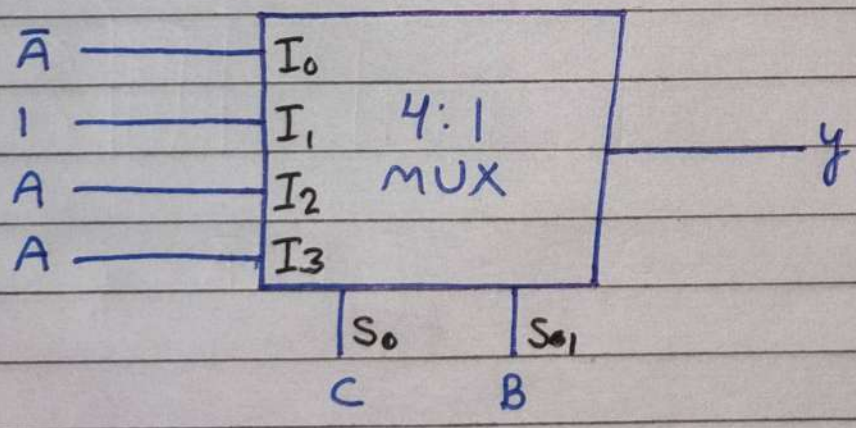
[Not in Gray Code Ordering.]

Trick: If ~~both~~ two marked in a column then one 1, if single marked in a column then the corresponding input line value and if zero marked in a column then zero 0.



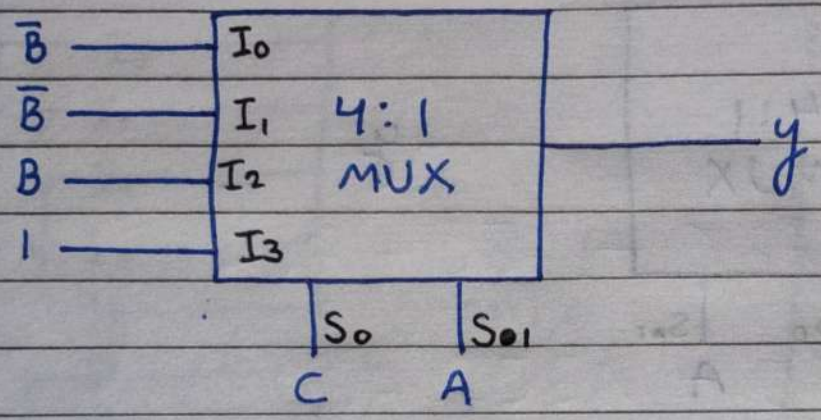
For B & C as select lines and A as input line,

	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	0	1	2	3
A	4	5	6	7
	$\bar{A}$	1	A	A

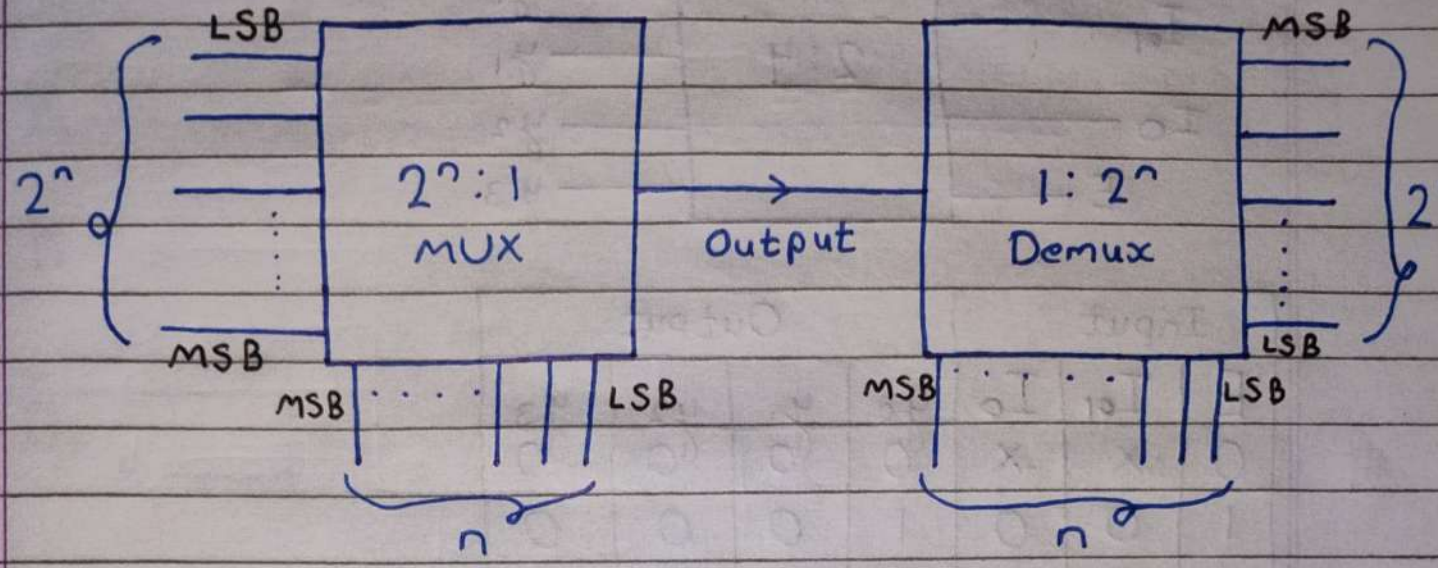


For A & C as select lines and B as input line,

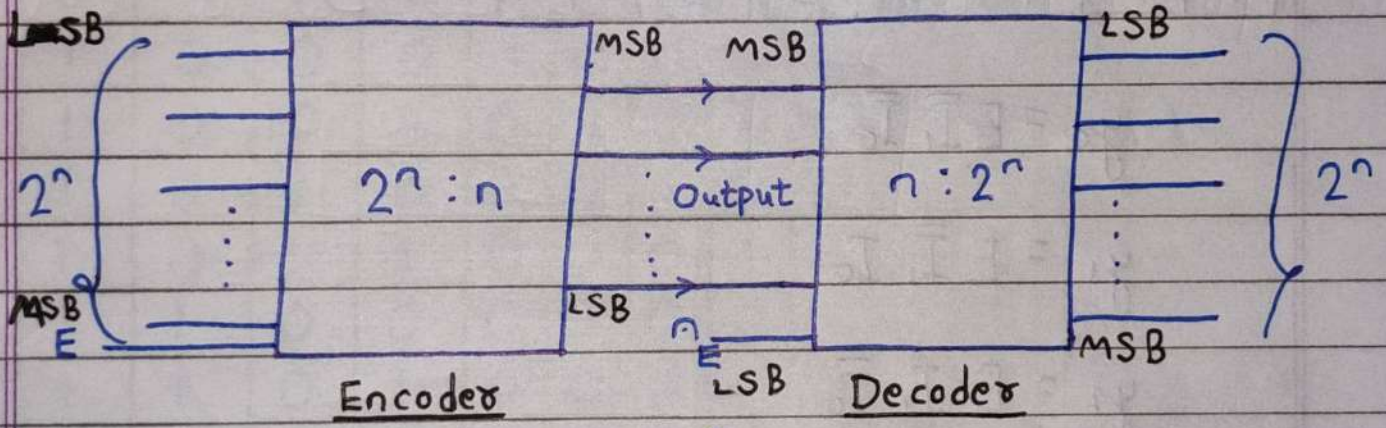
	$\bar{A}\bar{C}$	$\bar{A}C$	$A\bar{C}$	AC
$\bar{B}$	0	1	4	5
B	2	3	6	7
	$\bar{B}$	$\bar{B}$	B	1



De-multiplexer (DEMUX)



Encoder & Decoder



conversion of

Uses: Used in binary to octal, hexadecimal, BCD, Gray code, etc.

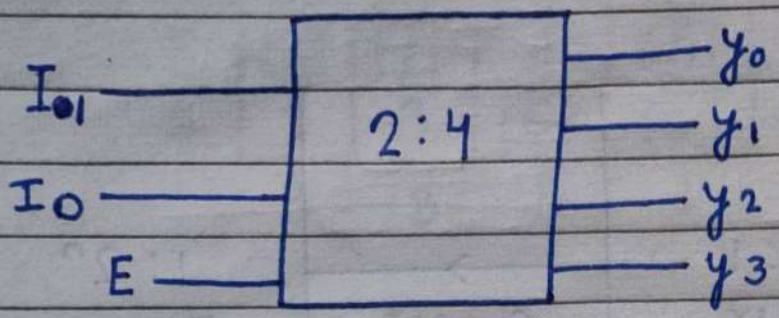
& decoder

In encoder, there is an extra pin E which is used to make encoder active or inactive.

$E = 0 \Rightarrow POS$

$E = 1 \Rightarrow SOP$

2:4 Decoder



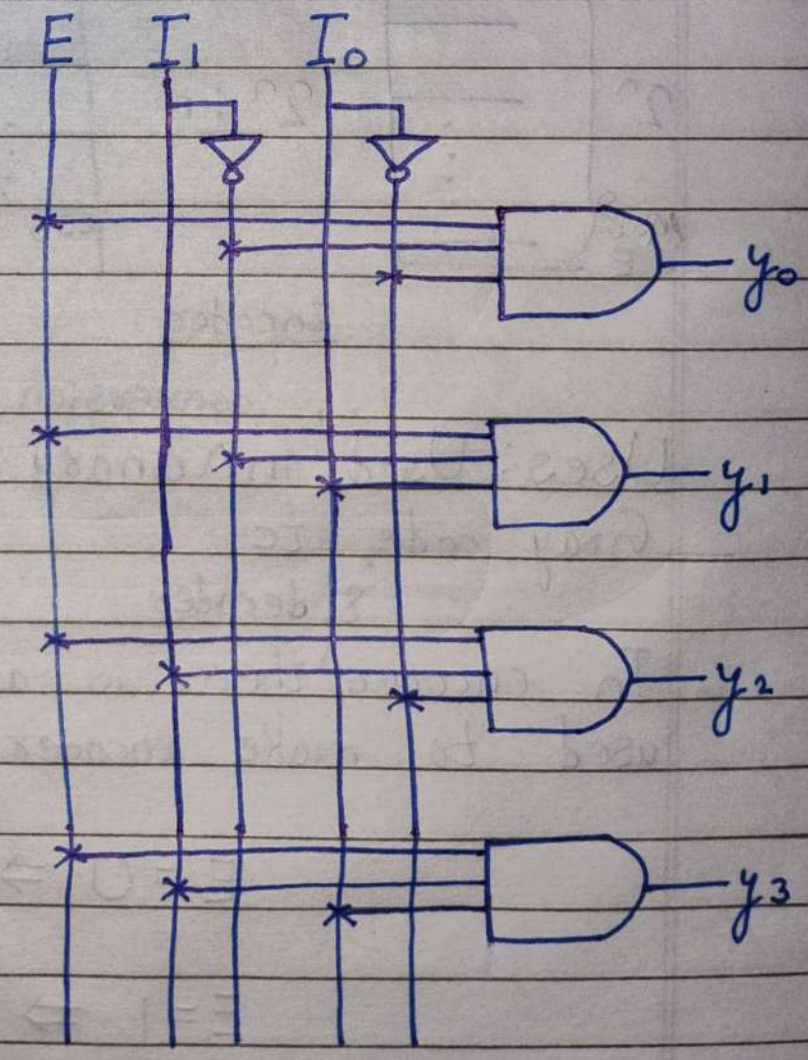
Input			Output			
E	I_1	I_0	y_0	y_1	y_2	y_3
0	x	x	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

$y_0 = E \bar{I}_1 \bar{I}_0$

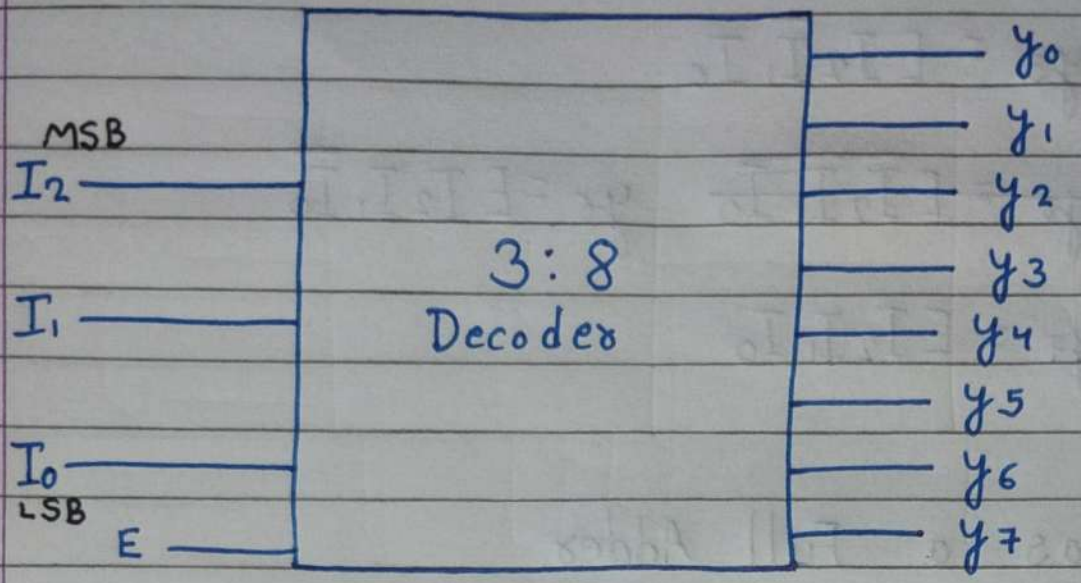
$y_1 = E \bar{I}_1 I_0$

$y_2 = E I_1 \bar{I}_0$

$y_3 = E I_1 I_0$



3:8 Decoder



Input				Output							
E	I_2	I_1	I_0	y_0	y_1	y_2	y_3	y_4	y_5	y_6	y_7
0	X	X	X	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	1

$$y_0 = E, \bar{I}_2 \bar{I}_1 \bar{I}_0$$

$$y_1 = E \bar{I}_2 \bar{I}_1 I_0$$

$$y_2 = E \bar{I}_2 I_1 \bar{I}_0$$

$$y_3 = E \bar{I}_2 I_1 I_0$$

$y_4 = E I_2 \bar{I}_1 \bar{I}_0$

$y_5 = E I_2 \bar{I}_1 I_0$

~~$y_6 = \overline{E I_2 I_1 I_0}$~~ $y_6 = E I_2 I_1 \bar{I}_0$

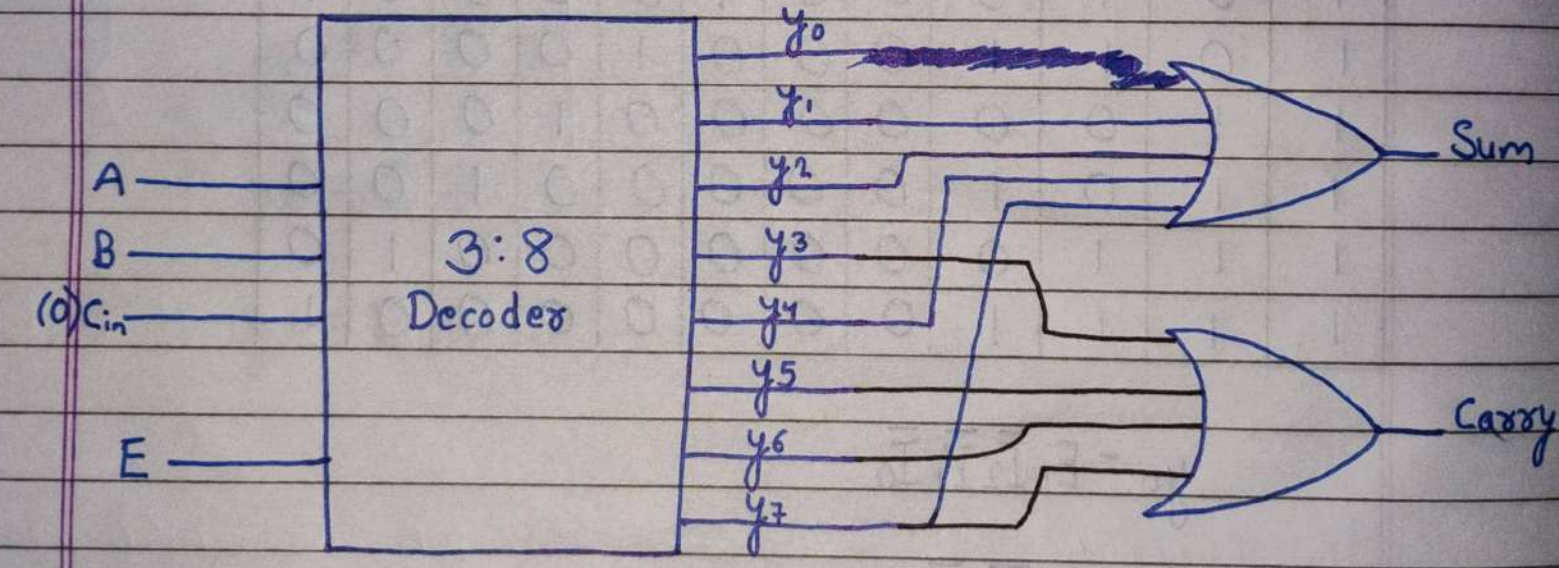
$y_7 = E I_2 I_1 I_0$

Decoder as a Full Adder

For full adder, we have,

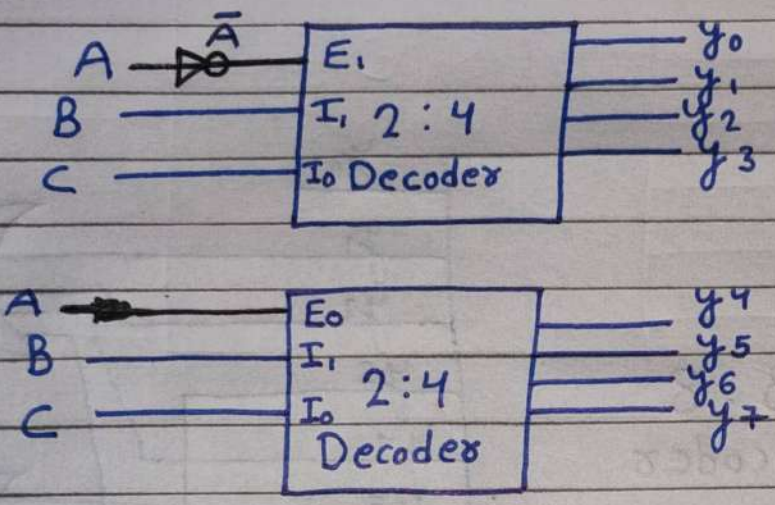
$Sum = \Sigma_m (1, 2, 4, 7)$

$Carry = \Sigma_m (3, 5, 6, 7)$



Q.1) Implement 3:8 (or 3x8) decoder using 2:4 decoder?

Soln:



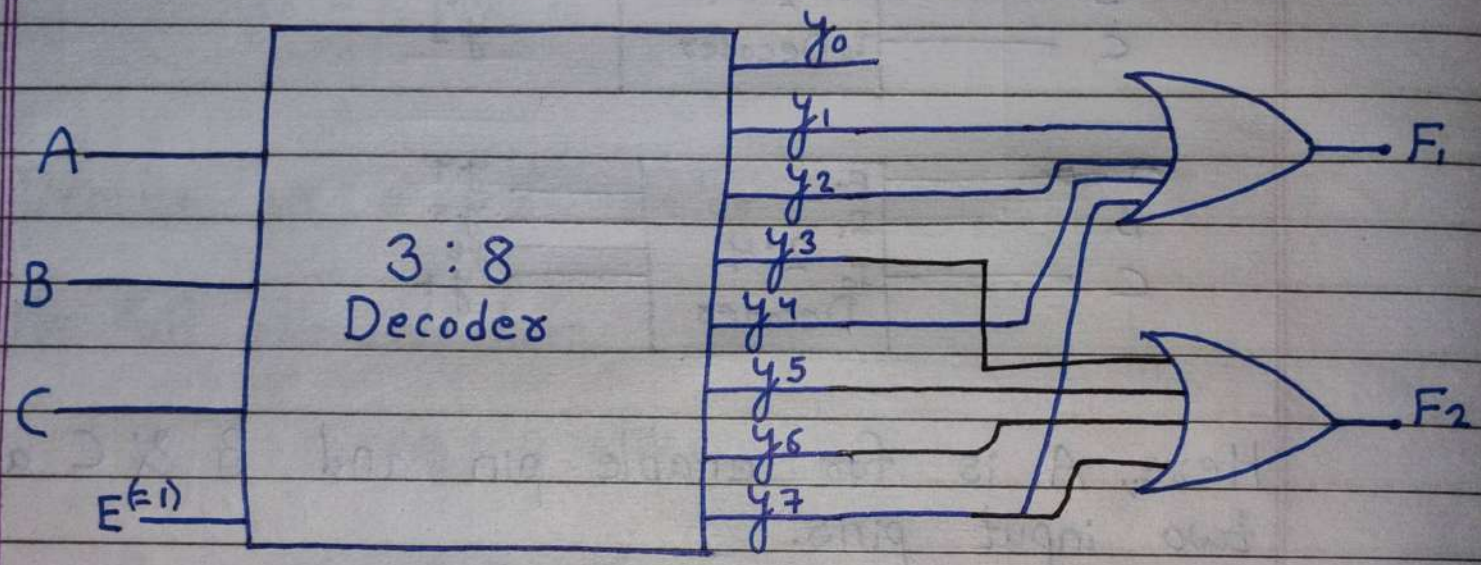
Here, A is for enable pin and B & C are two input pins.

A	B	C	E_0	E_1	Output
0	0	0	0	1	y_0
0	0	1	0	1	y_1
0	1	0	0	1	y_2
0	1	1	0	1	y_3
1	0	0	1	0	y_4
1	0	1	1	0	y_5
1	1	0	1	0	y_6
1	1	1	1	0	y_7

Q.2) Implement the given boolean expression using decoder.

$$F_1(A, B, C) = \Sigma_m (1, 2, 4, 7)$$
& $$F_2(A, B, C) = \Sigma_m (3, 5, 6, 7)$$

Soln:



More Topics Remaining